L25 The Taylor Polynomial Approximation

Section 8.1

Def. The Taylor Polynomial of degree n centered at x_0 approximating $f(x)$ with n derivatives at x_0 is given by

center

Taylor polynomial

$$p_n(x) = f(x_0) + f'(x_0)(x - x_0) + \frac{f''(x_0)}{2!}(x - x_0)^2$$

$$+\frac{f'''(x_0)}{3!}(x - x_0)^3 + \ldots + \frac{f^{(n)}(x_0)}{n!}(x - x_0)^n$$

$$= \sum_{k=0}^{n} \frac{f^{(k)}(x_0)}{k!}(x - x_0)^k$$

$0, 1, 2, \ldots, n$

index

ex. Determine the fourth-degree Taylor Polynomial for $\cos x$ at $x_0 = \pi/2$

$k = 0, 1, 2, 3, 4$

$f(x)$

$$P_4(x) = \sum_{k=0}^{4} \frac{f^{(k)}(\pi/2)}{k!}\left(x - \frac{\pi}{2}\right)^k = f\left(\tfrac{\pi}{2}\right) + f'\left(\tfrac{\pi}{2}\right)\left(x - \tfrac{\pi}{2}\right) + \ldots + \frac{f^{(4)}\left(\frac{\pi}{2}\right)}{4!}\left(x - \tfrac{\pi}{2}\right)^4$$

$n = 4$

$f(x) = \cos(x)$ $\quad f\left(\tfrac{\pi}{2}\right) = 0$

$f'(x) = -\sin(x)$ $\quad -1$

$f''(x) = -\cos(x)$ at $x = \tfrac{\pi}{2}$ $\quad 0$

$f'''(x) = \sin(x)$ $\quad 1$

$f^{(4)}(x) = \cos(x)$ $\quad f^{(4)}\left(\tfrac{\pi}{2}\right) = 0$

$$= \cancel{0} + (-1)\left(x - \tfrac{\pi}{2}\right) + \cancel{\frac{0}{2!}\left(x - \tfrac{\pi}{2}\right)^2}$$

$$+ \frac{1}{3!}\left(x - \tfrac{\pi}{2}\right)^3 + \cancel{\frac{0}{4!}\left(x - \tfrac{\pi}{2}\right)^4}$$

$$= -\left(x - \tfrac{\pi}{2}\right) + \frac{1}{6}\left(x - \tfrac{\pi}{2}\right)^3$$

ex. Solve $y' = y^2, \quad y(0) = 2$.

separable

$$\frac{dy}{dx} = y^2 \rightarrow \int \frac{1}{y^2}\,dy = \int dx$$

$$\Rightarrow \; -\frac{1}{y} = x + C$$

$$\Rightarrow \; y = \frac{-1}{x + C}$$

$$y(0) = 2 \Rightarrow \quad 2 = y(0) = \frac{-1}{0 + C} \Rightarrow 2 = \frac{-1}{C} \Rightarrow C = -1/2$$

$$\text{so}, \quad y = \frac{-1}{x - \frac{1}{2}}$$

<u>Recall</u>: $\dfrac{1}{1-x} = \displaystyle\sum_{n=0}^{\infty} x^n \quad , -1 < x < 1$

$$\frac{-1}{x - \frac{1}{2}} = \frac{-1}{-\frac{1}{2}(1 - 2x)} = 2 \cdot \frac{1}{1 - 2x}$$

$$= 2 \cdot \sum_{n=0}^{\infty} (2x)^n$$

$$= 2 \cdot \sum_{n=0}^{\infty} 2^n x^n = \sum_{n=0}^{\infty} 2^{n+1} x^n$$

$$= 2 + 4x + 8x^2 + 16x^3 + \cdots$$

ex. Find the 3rd degree Taylor Polynomial about $x_0 = 0$ that approximates the solution to the ODE. for $y(x)$ center

$$y' = y^2, \quad y(0) = 2.$$

Need $\underset{=2}{y(0)}$, $\underset{=4}{y'(0)}$, $\underset{=16}{y''(0)}$, and $\underset{=96}{y'''(0)}$.

use the eqn to find the other derivatives

$y' = y^2 \Rightarrow y'(0) = [y(0)]^2 = 2^2 = 4$

For y'' $\Rightarrow (y')' = (y^2)'$

$\Rightarrow y'' = 2yy'$

$\Rightarrow y''(0) = 2y(0)y'(0) = 2(2)(4) = 16$

y''': $(y'')' = (2yy')'$

$\Rightarrow y''' = (2y')y' + 2y(y'')$

$\Rightarrow y'''(0) = (2[y'(0)]^2) + 2y(0)y''(0)$

$= 2[4]^2 + 2(2)(16) = 32 + 64 = 96$

So $P_3(x) = y(0) + y'(0)x + \frac{y''(0)}{2!}x^2 + \frac{y'''(0)}{3!}x^3$

$= 2 + 4x + \frac{16}{2}x^2 + \frac{96}{6}x^3$

$= 2 + 4x + 8x^2 + 16x^3$

ex. Find the Taylor Polynomial for $f(x) = \frac{2}{1-2x}$

about $x_0 = 0$.

See previous

ex. Find the 4th degree Taylor Polynomial approximating the solution about $x_0 = 0$.

$y'' + y = e^{2x}, \quad y(0) = 0, \quad , y'(0) = 1.$

$y''(0) = 1$
$y'''(0) = 1$
$y^{(4)}(0) = 3$

solve for y'': $y'' = e^{2x} - y$

$y''(0)$: $y''(0) = e^{2(0)} - y(0) = 1 - 0 = 1$

y''': $(y'')' = (e^{2x} - y)' \Rightarrow y''' = 2e^{2x} - y'$

$y'''(0)$: $y'''(0) = 2e^{2(0)} - y'(0) = 2 - 1 = 1$

$y^{(4)}$: $y^{(4)} = (2e^{2x} - y')' = 4e^{2x} - y''$

$y^{(4)}(0)$: $y^{(4)}(0) = 4e^{2(0)} - y''(0) = 4 - 1 = 3$

So $P_4(x) = 0 + 1 \cdot x + \frac{1}{2!}x^2 + \frac{1}{3!}x^3 + \frac{3}{4!}x^4$

$= x + \frac{1}{2}x^2 + \frac{1}{6}x^3 + \frac{3}{24}x^4$

$= x + \frac{1}{2}x^2 + \frac{1}{6}x^3 + \frac{1}{8}x^4$

ex. Find the exact solution.

$$y'' + y = e^{2x}, \quad y(0) = 0, \quad , y'(0) = 1.$$

Laplace: $(s^2 Y(s) - s\cancel{y(0)}^{\,0} - \cancel{y'(0)}^{\,1}) + Y(s) = \frac{1}{s-2}$

$\Rightarrow (s^2+1)Y(s) - 1 = \frac{1}{s-2}$

$\Rightarrow (s^2+1)Y(s) = 1 + \frac{1}{s-2}$

$\Rightarrow Y(s) = \frac{1}{s^2+1} + \frac{1}{(s-2)(s^2+1)}$

$\frac{1}{(s-2)(s^2+1)} \overset{PFD}{=} \frac{A}{s-2} + \frac{Bs+C}{s^2+1} \Rightarrow \begin{array}{l} A = \frac{1}{5} \\ B = -\frac{1}{5} \\ C = -\frac{2}{5} \end{array}$

$\Rightarrow Y(s) = \frac{1}{s^2+1} + \left(\frac{1/5}{s-2} + \frac{-\frac{1}{5}s - \frac{2}{5}}{s^2+1} \right)$

$\Rightarrow y'(x) = \sin(x) + \frac{1}{5}e^{2x} - \frac{1}{5}\cos(x) - \frac{2}{5}\sin(x)$

$= \frac{1}{5}e^{2x} - \frac{1}{5}\cos(x) + \frac{3}{5}\sin(x)$

Recall: $e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}$

$\sin(x) = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} x^{2n+1}$

$\cos(x) = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!} x^{2n}$

$$\frac{1}{5}e^{2x} = \frac{1}{5}\sum_{n=0}^{\infty}\frac{(2x)^n}{n!} = \frac{1}{5}\left(1+2x+\frac{4x^2}{2}+\frac{8x^3}{6}+\frac{16x^4}{24}+\cdots\right)$$

$$-\frac{1}{5}\cos(x) = -\frac{1}{5}\left(1-\frac{x^2}{2}+\frac{x^4}{24}-\frac{x^6}{6!}+\cdots\right)$$

$$+\frac{3}{5}\sin(x) = \frac{3}{5}\left(x-\frac{x^3}{6}+\frac{x^5}{5!}-\cdots\right)$$

$$= 0+\left(\frac{2x}{5}+\frac{3x}{5}\right)+\left(\frac{4}{10}x^2+\frac{x^2}{10}\right)+\left(\frac{8}{30}x^3-\frac{3}{30}x^3\right)$$

$$+\left(\frac{16}{120}x^4-\frac{1}{120}x^4\right)+\cdots$$

$$= 0+x+\frac{x^2}{2}+\frac{x^3}{6}+\frac{x^4}{8}+\cdots$$

ex. Find the Taylor Polynomial approximation around $\underline{\underline{x_0}}$ $= 0$ for the initial value problem.

$$y'' = 3y' + x^{7/3}y, \quad y(0) = 10, \quad y'(0) = 5.$$

$y''(0) = 3y'(0) + (0)^{7/3}\, y(0) = 3(5) + 0 = 15$

$y''' : \quad y''' = (3y' + x^{7/3}y)' = 3y'' + \frac{7}{3}x^{4/3}y + x^{7/3}y'$

$\Rightarrow y'''(0) = 3y''(0) + \frac{7}{3}0^{4/3}y(0) + 0^{7/3}\,y'(0) = 45$

$y^{(4)}: \quad y^{(4)} = (3y'' + \frac{7}{3}x^{4/3}y + x^{7/3}y')'$

$\Rightarrow y^{(4)} = 3y''' + (\frac{7}{3}\cdot\frac{4}{3}x^{1/3}\,y + \frac{7}{3}x^{4/3}y') + (\frac{7}{3}x^{4/3}y' + x^{7/3}y'')$

$\Rightarrow y^{(4)}(0) = 135$

$y^{(5)}: \quad y^{(5)} = 3y^{(4)} + (\frac{7}{3}\cdot\frac{4}{3}\cdot\frac{1}{3}x^{-2/3}y + \frac{7}{3}\cdot\frac{4}{3}x^{1/3}y')$

$+ (\frac{14}{3}\cdot\frac{4}{3}x^{1/3}y' + \frac{14}{3}x^{4/3}y'')$

$+ (\frac{7}{3}x^{4/3}y'' + x^{7/3}y''')$

$y^{(5)}(0) =$ not defined!

We can use $p_4(x) = 10 + 5x + \frac{15}{2}x^2 + \frac{45}{6}x^3 + \frac{135}{24}x^4$

but no higher degree taylor polynomial $\underline{\underline{\text{at}}}$ $\underline{\underline{x=0}}$.

ex. Find the Taylor Polynomial of degree 3 around x_0 for the initial value problem.

$$y' = \frac{1}{x+y+1}, \quad y(0) = 0.$$

$y'(0) = 1$

$y''(0) = -2$

$y'''(0) = 10$

$$y'(0) = \frac{1}{0+0+1} = 1$$

$$y'' = \left((x+y+1)^{-1}\right)' = -(x+y+1)^{-2}(1+y')$$

$$y''(0) = -(0+0+1)^{-2}(1+1) = -2$$

$$y''' = \left(-(x+y+1)^{-2}(1+y')\right)' = 2(x+y+1)^{-3}(1+y')(1+y') + \left(-(x+y+1)^{-2}(y'')\right)$$

$$y'''(0) = 2(0+0+1)^{-3}(1+1)^2 + \left(-(0+0+1)^{-2}(-2)\right)$$

$$= 8 + 2 = 10$$

$$P_3(x) = 0 + 1 \cdot x + \frac{-2}{2}x^2 + \frac{10}{6}x^3$$

$$= x - x^2 + \frac{5}{3}x^3$$ ← approximates an explicit solution.